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Delivering Operational Excellence in Mature Oil & Gas Assets

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Mature upstream oil & gas businesses are facing challenges to performance across multiple fronts. Upstream assets in late life, in regions with complex subsurface conditions, or in difficult operating environments can face a perfect storm of production decline, rising unit costs, fiscal pressures, and trade-offs around value. The most common response across the industry has been to seek opex and capex cuts through supplier cost renegotiations, headcount reductions, and by limiting spending on exploration, projects, and even asset integrity. These efforts usually deliver savings in the short term, but the outcomes are often not sustainable, and the opportunity cost of foregoing attractive investments can be high.

This paper describes a better way, avoiding complex, short-term cost reductions and instead mobilizing resources to deliver more impactful and lasting improvements in overall performance and value. This approach is more motivating and exciting for executives and teams who are exhausted by repeated cost cutting efforts, and delivers much greater value over a longer period. The approach is very much different from continuous improvement, as it still requires the mobilization of organizations and capabilities to deliver, and needs effective change management. But the prize for getting this right is substantial.

Introduction

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Industry Challenges

KPMG has seen a common pattern driving concerns about the viability among upstream operators in both International Oil Companies (IOCs) and National Oil Companies (NOCs):

Production decline and system balancing challenges

Continuing falls in oil & gas production from many maturing assets, and challenges in many regions and areas on reserve replacement and maintaining commitments to lenders for reserve-based lending. In resource-rich areas, related challenges in trying to expand production in a way that doesn't drive up unit cost

Limited capital availability and challenges around ensuring that modifications and growth investments are fit for purpose

Increasingly tight fiscal challenges, and strong expectations that upstream investments will have superior IRR performance and be resilient to lower pricing

Pressure on resources and capabilities

Continued generational/succession issues, and losses of key personnel in many regions as decommissioning accelerates

Unit cost and asset reliability issues

Aging assets and limited resources leading to challenges around wells and processing asset availability and rising opex/boe

Left unchecked, these pressures could cause companies to reduce investments in assets that appear uncompetitive, potentially even leading to early decommissioning or sale, on the assumption that core assets are simply not competitive. But that path feels deeply unattractive for most operators, who believe that it ought to be possible to turn things around.

These converging issues are causing upstream operators to seek bold, fresh methods for unlocking performance, cutting decline rates, reducing costs, and supporting growth.

Traditional Responses: Why the Old Playbook Falls Short

How have operators responded to these challenges? All too often, we've observed reliance on a familiar playbook that focuses overwhelmingly on top-down operating cost reductions and capital rationing. Across the industry, that playbook has centered on squeezing suppliers on both goods and services, reducing offshore and onshore manning, outsourcing support functions and transactional activities, and squeezing spending on integrity management, maintenance, and engineering support. These measures are often combined with large numbers of continuous improvement efforts that deliver short-term gains, but don't genuinely move the needle on performance, along with capex reductions through cuts to exploration and drilling activities and the deferral or cancellation of spending on facilities upgrades and new developments.

Programs like these often prove both intensive and exhausting, and in many cases the savings they achieve prove ephemeral, leading to repeated waves of cost cutting across the same assets.

In fact, we have often seen knock-on effects from such efforts that worsen performance and potentially even create a vicious circle of continuing cost cutting that accelerates underperformance, creating more pressure for further cost reductions.

For example, owners of individual cost reduction initiatives are often incentivized to optimize delivery in isolation, potentially sabotaging the overall objectives of other initiatives. We've observed cases in which a misapplication of maintenance cost benchmarks leads to targets to cut maintenance cost by 10% or more. Teams then focus on this cost target, reducing the number of technicians by carrying out less preventative maintenance. The outcome is that maintenance staff cost falls, but the reduced preventative maintenance effort has knock-on impacts on reliability and facilities integrity issues worsen, leading to both safety problems and the needs for increased expenditure down the road.

Similarly, some cost reduction efforts are drawn from techniques taken from companies undergoing formal restructuring processes, with the enormous and continuous top-down pressure on technical teams who try to deliver their targets on top of their daily responsibilities. As a result, initiatives meet formal deadlines by sacrificing quality and depth, and teams end up exhausted by constant sprint meetings and reporting.

A Call for Change: The New Playbook

A different approach is clearly needed. We have witnessed a growing recognition from industry leaders that even in the "traditional" business, cost cutting alone won't be sufficient to cope with emerging challenges. We see five enduring issues that cannot be addressed by spending reduction alone (see figure 1).



Figure 1—Challenges that require measures beyond traditional cost reductions

This is why the sophisticated upstream players are beginning to look beyond cost-led efforts and are instead shifting towards combined cost and performance programs, focused on value creation in both legacy assets and new businesses.

But this is not an easy process, as getting it right means looking at opportunities in a new and different way, and—critically—it means overcoming a range of assumptions and behaviors among both senior leaders and mid-level technical and commercial teams that can stand in the way of lasting change.

Teams who have been through repeated waves of opex reductions often become both cynical and resistant to any new approach. Almost any idea can be met with a weary account of how something similar was attempted years ago and failed, and so there is little use in trying again. We typically describe this resistance to deeper change in terms of three bad habits that reduce the appetite for looking beyond today's approaches:

Complacency. 'We've got it covered: There's no additional value to be unlocked beyond what we're doing already'

Complexity. 'The area is too complex to be able to generate a conclusion that justifies any change'

Risk Aversion. 'The potential value of any opportunity does not justify even a theoretical increase in any existing approach to managing risk'

Overcoming these barriers is critical to successfully finding and unlocking hidden value, but the starting point is to look for value in a different way, reaching beyond the most familiar approaches, and being prepared to ask bigger questions about how businesses operate and deliver.

This paper describes the most promising areas where KPMG has repeatedly identified material and tangible value gains and—crucially—discusses a different approach to organizing around value identification and delivery.

What a Different Approach Means in Practice

Our experience shows that something different is needed to deliver material and lasting value from mature upstream assets.

Five elements form the core of the approach:

A change in focus from cost to **unit cost**—with more focus on production and performance KPIs vs. budgetary KPIs. In other words, we believe that financial benefits are best captured as outcomes from improving performance, rather than leading with budget cuts

Limited but intensive campaigns around fundamental drivers of performance: substantial ambition and resourcing to deliver change that looks and feels fundamentally different from continuous improvement

Less is more: A few comprehensive and targeted initiatives rather than dozens or hundreds of small efforts

Seeking to address bottlenecks in **activity levels and business complexity**, reducing underlying pressure on cost and resource inflation

A willingness to **challenge the fundamentals of the business operating model**. Seeking to act like a large independent rather than a complex integrated player

Actively applying **new technologies** (both physical technologies and especially digital and AI) as enablers to performance, but not as something to be pursued for its own sake

At the heart of these efforts is the need to target specific areas of operational performance that combine the highest sustainable materiality and unit cost potential with ties to reductions in business complexity and costly models for technical support and business complexity.

What are the five "big hit" opportunity areas within mature asset upstream businesses?



Figure 2—Five Transformational Opportunity Areas

Operational excellence transformation

Across multiple assets, clients and regions, we have seen some common patterns:

- Significant unaccounted losses impacting reliability and availability
- Ineffective maintenance delivery efficiency and low time on tools
- Data complexity and quality standing in the way of effective planning and decision making

Tackling these issues requires integrated efforts that have two core elements:

1. A fundamental focus on driving improved availability and reliability at lower cost, through a changed approach to criticality assessment, maintenance strategies and maintenance delivery models.

Increased production from reliability gains has far better economics than efforts requiring capital investments in new wells or facilities expansion

2. Driving improved drilling and well intervention performance, focusing on new arrangements to drive down non-productive time, assess and value well intervention opportunities appropriately and adopt new delivery models

Capital delivery and turnaround effectiveness

There is a common story about the barriers to effective capital delivery:

- Variability and uncertainty on returns due to fluctuating oil / gas market prices
- Lengthy delays (40-80%) in delivery times for major projects, eroding expected returns
- Turnarounds typically overscoped, and exceeding planned durations

Top-down capital budget rationing won't solve these. Instead, we prefer to focus on

1. *Driving complexity reduction in project portfolio management and opportunity selection*
2. *Achieving step change in delivery effectiveness through better planning, aligned contractor incentives, and a bias against customization and gold plating. Improving turnaround performance is often a big upside area. Reducing turnaround frequencies and rigorously limiting scope to shorter turnaround durations can lead to a huge production upside*

De-risking supply chain constraints. *Demands to suppliers for arbitrary rate reductions don't address the root causes of supply chain inefficiency:*

- *Growing capacity and capability constraints*
- *Potential of AI on contract management and buying not realized*
- *Traditional tendering and supplier management creates inefficiency and builds structural costs into assets*

We have seen the greatest upsides across two areas:

1. *Aligning procurement strategy with asset strategies, with a greater focus on standardization and modularization*
2. *Leveraging scale, and buyer power, category strategies and technology with a focus on reductions in supplier numbers, increases in scale, and conscious sharing of upsides. From this point of view, the typical approach of demanding a one-off 10% rate reduction from suppliers looks less like effective negotiating, and more like poor supplier management*

Organizational capacity and capability. *Large oil & gas companies are simply too complex. Although the largest operators maintain their faith in building centralized capabilities and highly standardized delivery models, we observe a continuing pattern in which smaller independents outperform their bigger rivals.*

Common approaches often fail to address the real issues:

- *Capabilities and operating models not keeping up with pace of change, creating gaps in execution and failure to "cash the cheque" on opportunities*
- *Tension among regional operators on how to drive operational efficiency while maintaining mandated localization targets*
- *Large and complex central delivery functions that deliver increases in overhead cost without realizing scale advantages*

In other words, large players can become focused on delivering complex services and activities more effectively, in situations where their smaller peers wouldn't see the need for those activities in the first place. Overcoming this means two things in practice, thinking like independent operators:

1. *Evolving operating models to build and redeploy capabilities at lower cost and complexity, potentially eliminating whole swathes of central functions and services and moving towards ultra-lean delivery models for technical, commercial and back office functions, rather than relying on outsourcing or process redesign to while chasing a prize that never gets realized.*
2. Supercharge strategic workforce planning, focusing on challenging views on required resourcing and accessing skills rather than just identifying and remedying resource gaps. Paradoxically, the standard cost-cutting approach of offering early retirements to the most experienced staff in the company can reduce productivity and effectiveness if a strategic view isn't applied.

Technology and Artificial Intelligence (AI)

Two broad positive trends are in evidence today:

- AI is gaining traction and being used to optimize activities across the upstream operations including exploration, drilling, maintenance, decarbonization etc.
- Operators are investing in new technologies which they believe will give them competitive edge by automation and re-training employees into more value-adding activities

But these are often offset by a much more damaging third trend: a tendency to treat AI as a panacea. No matter what the issue is, some companies see AI as the answer. At times, the outcome can almost look farcical, in which teams come under pressure to generate AI-led initiatives without any consideration about whether they will truly add value. We saw the same tendency in play a few years earlier, when the best way to gain central funding for almost any initiative was to put the word "digital" in front of it.

This doesn't by any means suggest that AI has no benefits. We believe in the potential of AI but think that there is a better way to go about AI identification and deployment than today's "AI everywhere and in everything" approach. That means:

1. More targeted efforts that address real areas of shortfalls, starting with performance and production needs rather than treating technology efforts like an independent value driver
2. Accelerated AI deployment, with focus on 'returns' and impact on assets rather than chasing technology. In general, we believe that if a program cannot be directly or indirectly valued either financially or in terms of a related metric, it is most likely not a good investment

Why a program following these principles feels different from typical efforts

What does it mean for an upstream company to avoid a typical opex-led, granular opportunity approach?

First and foremost, it means that efforts get organized around simpler, more integrated initiatives covering the big hit items described above. Every action and activity in the efforts is aimed at delivering measurable changes to meaningful physical and financial performance measures that reflect the underlying value drivers of the business (see [figure 3](#)).

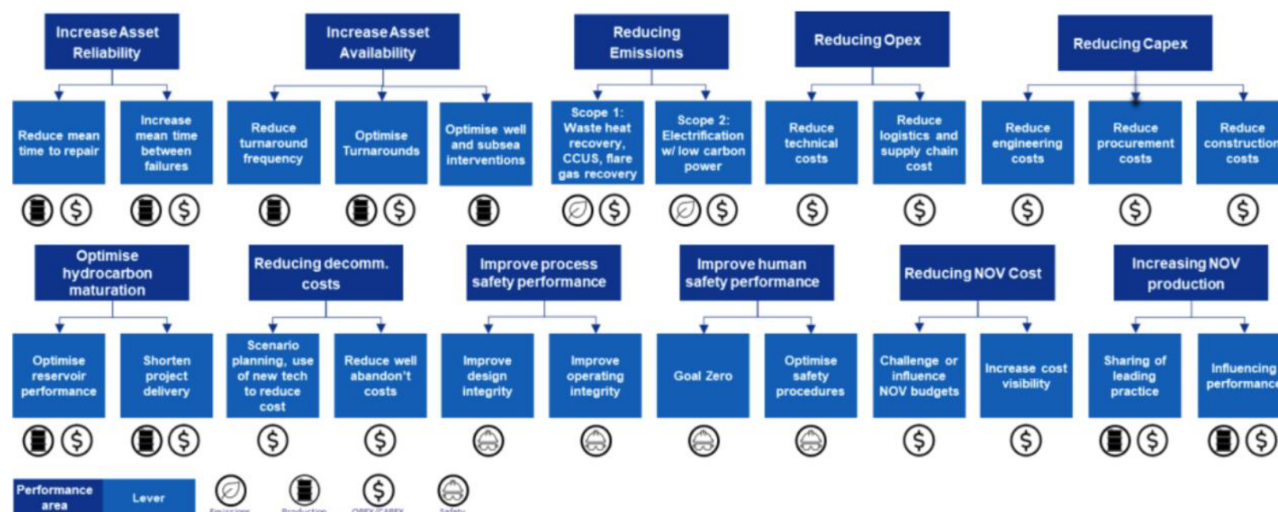


Figure 3—Key performance drivers to address in integrated program delivery

In practice, this means that delivery isn't about hundreds of initiatives managed through a consultant-supplied tracking tool; but about more comprehensive efforts with clearer leadership and responsibility, often with dedicated resources assigned for delivery. Integrated efforts have a lot of advantages. They can be better planned, more strategic, and pay greater attention to interfaces and reinforcing effects. And they are subject to less temptation to assume that delivery can be accomplished through sprints and side of desk efforts. For example, a reliability improvement program only works well if delivered through truly integrated efforts focusing on equipment strategies, maintenance planning, and maintenance delivery (see figure 4).

Improving asset reliability increases production, with the added benefit of opex reduction

Moving towards best in class reliability of >97% means maximising time between trips, while effectively returning assets to production when outages do happen. The prize is primarily about reduced deferrals, but with opex gains as well.



Figure 4—An example of a comprehensive effort to drive enhanced facilities reliability

We typically seek to target just 4-5 overall program verticals, with a common transformation management capability holding it together. In this case, the potential power and sustainability of the efforts can far exceed the outcomes from a more typical cost reduction effort: less truly is more.

How to start

Because success means focusing on fewer things with a more comprehensive and integrated approach, it is essential to focus on the right things.

We've learned some lessons on what that entails.

First and foremost, traditional benchmarking with normalized data sets and quartile positioning is rarely successful in setting the stage for the right kind of effort. This often ends up creating a theoretical gap to potential that raises interesting questions, but doesn't provide answers. Instead, we recommend very short, sharp up-front assessments that set tangible baselines, and then drive opportunity estimation through two major questions: what outcomes does strong performance deliver in reality? And what is it that comparators do in practice to realize these outcomes? (see [figure 5](#)).

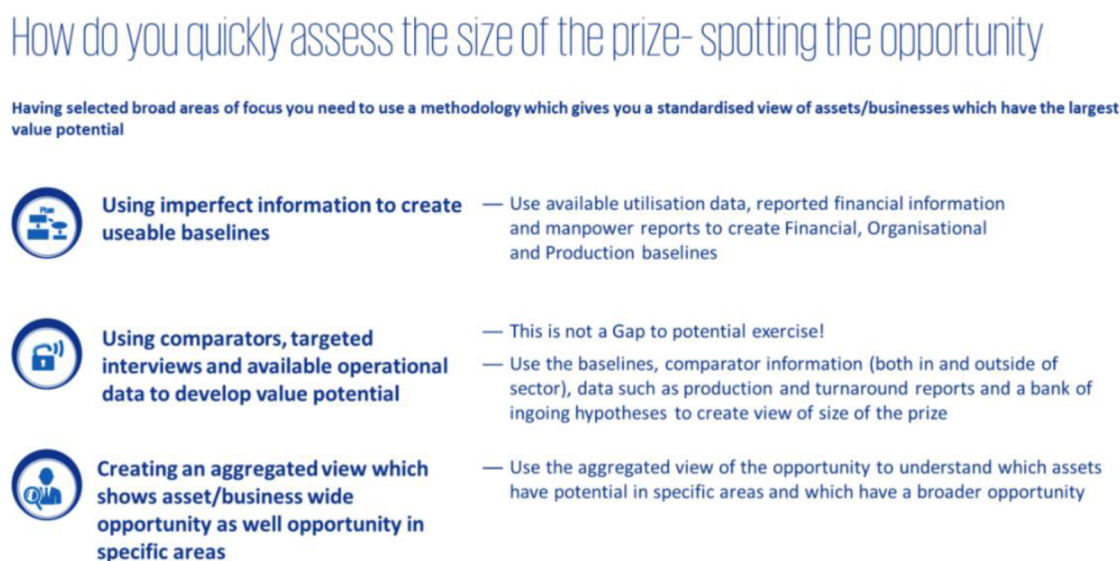


Figure 5—a pragmatic view of opportunity framing

The aim is to create a portfolio of major opportunity areas, rather than a long list of hundreds of individual initiatives that might make sense in isolation, but cannot possibly be delivered effectively in practice. This is especially the case where multiple independent initiatives create time and effort pressures on already stretched operations and support staff.

These efforts make it possible to decide which of the "big hit" opportunity areas have the most potential. In many cases, the most important choice is to NOT pursue a lower-value initiative that could prove a distraction.

A critical part of this effort is to take a realistic view of what resourcing and cost delivery will require. Optimism bias in shaping initiatives and opportunities is rife in many cases. We believe in stretch targets and ambition, but more transformative outcomes can rarely be achieved through hope alone.

These assessments are then followed by organizing a Transformation Management Office (TMO) structure and dedicated resourcing needed to deliver. We choose the term TMO consciously. Underfortunately, central Program Management Offices often do little more than organize reporting and tracking. But the deployment of respected and skilled individuals with credibility in the line and a willingness to roll up their sleeves can have a huge impact.

Holding it all together: people and leadership make all the difference

Change on any scale is hard, and although a comprehensive and integrated approach like that described here has a bigger and more lasting impact, the greatest predictor of success is the quality of the people involved,

a realistic view of what level of resourcing is required, and most of all, a leadership team that is determined to see things through.

We've synthesized our experience with successful transformations into "five golden rules." Efforts that follow these have a far greater chance of success than those that do not. Any falling short on any one of these can doom an otherwise well-intentioned effort (see [figure 6](#)).



Figure 6—The Five Golden Rules of successful upstream transformations

Conclusion

In summary, we believe that the right approach can be inspiring rather than exhausting. Inspirational rather than painful. And most of all, lasting in impact and outcomes.

Upstream oil & gas is a long-term business, in which carefully planned investments create high-value rewards that last over the many years, continuing to generate value and returns. We should ask nothing less from our efforts to deliver operational excellence in our mature oil & gas assets.